

Computing Methods for Physics – 26 June 2023

Your exam material (code files, plots, etc.) must be submitted via google classroom by 18:00 as a single zip file.

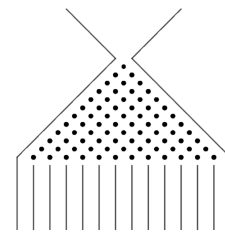
C++ evaluation will be based on: correct syntax, proper return types, proper arguments of functions, data members and class interfaces, unnecessary void functions, correct mathematical expressions, comments throughout the code, separation of class implementations and interfaces.

Python evaluation will be based on: correct syntax, avoiding C-style loops, using Python features in general, comments throughout the notebook/scripts, labels, legends and plot styling and clarity in general.

Part 1 – C++

Implement a class `GaltonBoard` to represent the usage of a Galton board with N rows.

The board is held vertically. Beads are dropped from the top single-bin row and bounce left or right on the pegs, with equal probability, while moving downwards from row to row. Each row has one more bin than the row immediately above it. At the end of their journeys, the beads are collected into the bins of the bottom row.



`GaltonBoard` must provide the following.

- A constructor to represent the board and track the number of experiments undertaken. Think of the board as an array with $1 + 2 + \dots + N = N(N + 1)/2$ elements (with N provided by the user) each of which is 0 when no bead has travelled.
- A method to simulate the evolution of a single bead from the top to the bottom: when the bead goes through a bin of the board, the array element corresponding to it increases by 1.
- A method to carry out the simulation of N_{exp} bead travels.
- A method to reset the array to 0 and lose memory of the simulations.
- A method to print to screen the content of the array so that the board rows are evident to the user.

Your submitted material must include a file `app.cpp` that showcases the class you implemented by simulating 3 experiments with 10, 100 and 1000 beads going through a 10-row Galton board.

Part 2 – Python

Use a Python notebook or Python scripts to complete the following tasks.

1. Design and implement a class called `mydataset` to store and manipulate N real values $\{d_0, d_1, \dots, d_{N-1}\}$.
2. Provide a constructor that fills in the d_i 's by drawing them from a Gaussian distribution centered in 0 with *default* standard deviation 1 (N is provided by the user).
3. Provide a method to plot the d_i 's: a first panel must display the values ordered by index, while a second panel must be a histogram of the distribution of values.
4. Provide a method that determines and stores the Discrete Fourier Transform:

$$D_k = \sum_{n=0}^{N-1} d_n e^{-i \frac{2\pi}{N} kn} \quad k \in \{0, 1, \dots, N-1\}.$$

5. Provide a method to plot the Discrete Fourier Transform: its three panels must show the amplitude of the D_k 's, their real parts, and their imaginary parts.